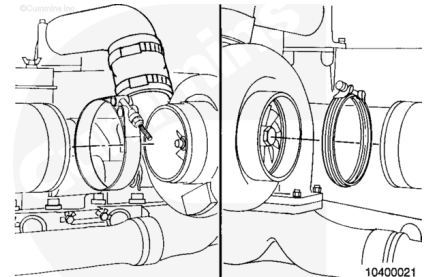


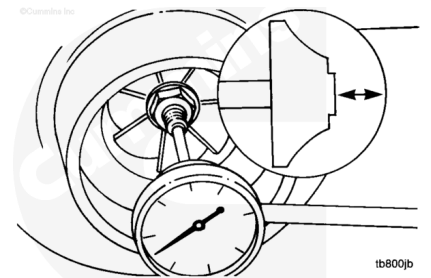
Trainings ▾

Initial Check

Remove the intake and exhaust piping from the turbocharger.



Use dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).



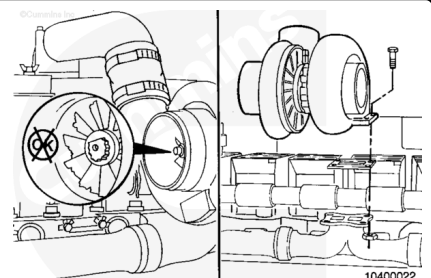
Compressor Inlet Axial Endplay			
	mm		in
Holset® HX82/HX8 3/HE800 Turbocharg er	0.025	MIN	0.001
	0.152	MAX	0.006

Make sure movement is within MIN/MAX Total Indicator Reading (TIR) values shown above.

Note : If the end clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Inspect the turbocharger compressor impeller blades for damage.

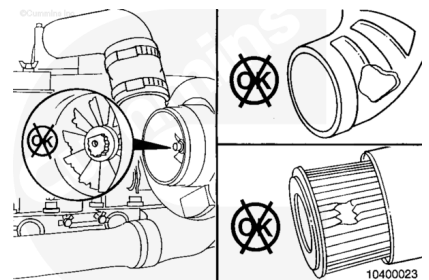
Rebuild or replace the turbocharger if damage is found.



If the impeller is damaged, inspect the intake air piping and the filter element for damage.

Inspect the compressor cover for any debris.

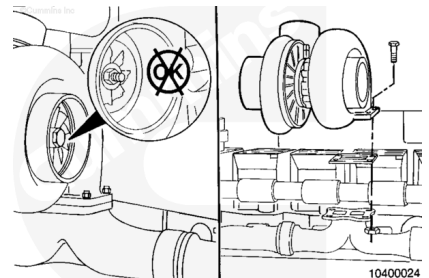
Repair any damage before operating the engine.



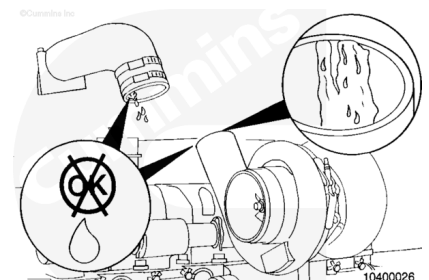
Inspect the turbine wheel for damage.

Check for debris stuck in the turbine housing.

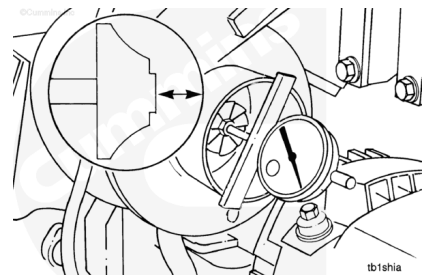
Replace or rebuild the turbocharger if damage is found.



Examine the compressor discharge and air crossover piping for oil.

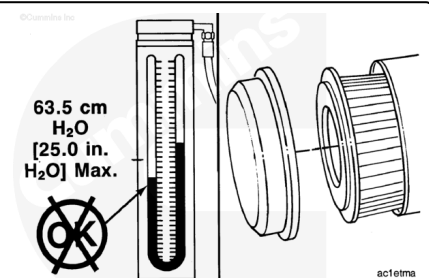


If oil is seen, check the axial motion and radial clearance of the turbocharger.

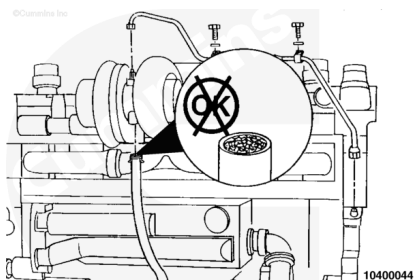


Check for intake restriction. Refer to Procedure 010-031 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-031.html>)

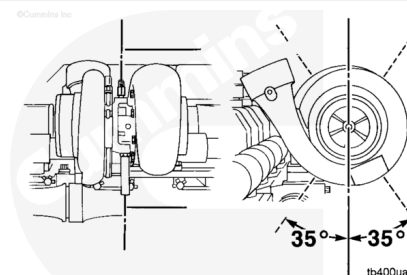
If the restriction exceeds specifications, replace or clean the air filter element. Refer to the equipment manufacture service information.



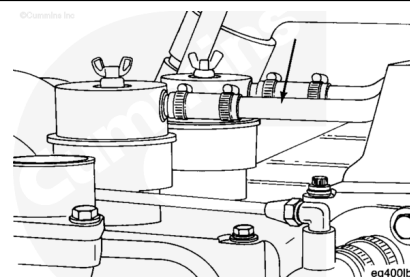
Remove the oil drain tube and check for restrictions. Clear any restrictions that are found.



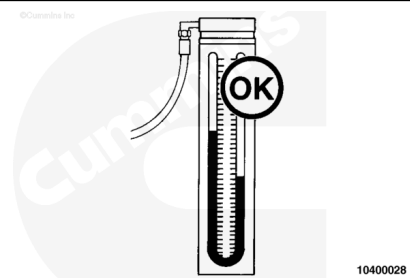
Check the angle of the drain tube. The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary.



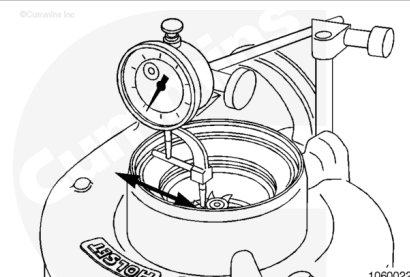
If the drain tube is free of restrictions and at the correct angle, check the crankcase breather tube for restriction. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html)



If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14. (/qs3/pubsys2/xml/en/procedures/20/20-014-005.html)



Measure the radial clearance (side-to-side) at the compressor impeller nose using a dial gauge.



Compressor Impeller Radial Endplay

	mm		in
Holset® HX82/HX8 3/HE800 Turbocharg er	0.254	MIN	0.01
	0.787	MAX	0.031

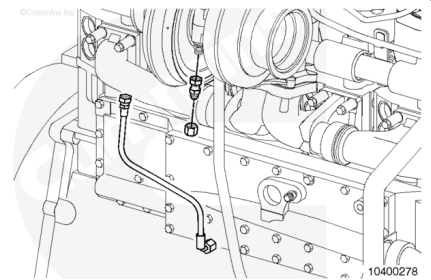
Make sure movement is within MIN/MAX Total Indicator Reading (TIR) values shown above.

Note : If the radial clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Leak Test

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)

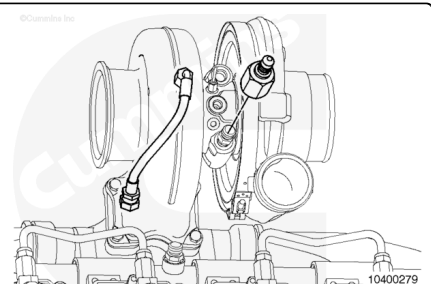
Remove the coolant supply hose or line from the turbocharger. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html>)

Install the reducing adapter and cap from the Turbocharger Coolant Leak Test Kit, Part Number 4918515.

Hand tighten the reducing adapter and cap.

Remove the turbocharger coolant return hose or line. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html>)

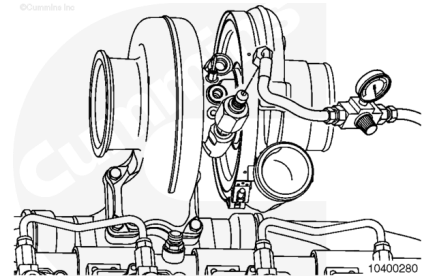
Install the reducing adapter from the Turbocharger Coolant Leak Test Kit, Part Number 4918515, on the turbocharger coolant return connection.



Hand tighten the connection.

Install the pressure regulator, Part Number 3164231, onto the coolant return reducing adapter.

Hand tighten the pressure regulator.



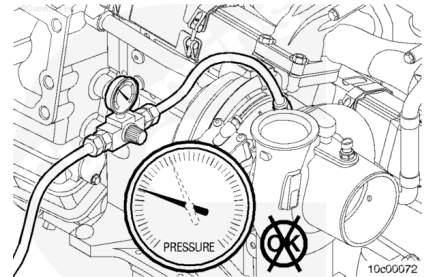
Connect shop air to the pressure regulator.

Pressurize the turbocharger to 276 kPa [40 psi].

Shut off the air pressure to the regulator.

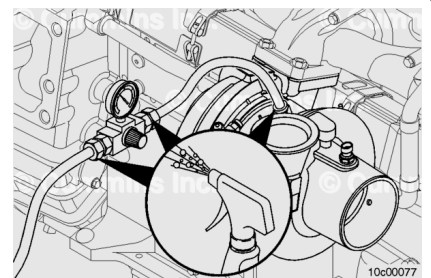
Watch for the pressure to decrease.

The air pressure **must not** decrease more than 49 kPa [5 psi] in 1 minute.



If the air pressure decreases greater than 49 kPa [5 psi] in 1 minute, use a spray bottle of soapy water to wet all air hose connections. Bubbles will appear if the connections are leaking.

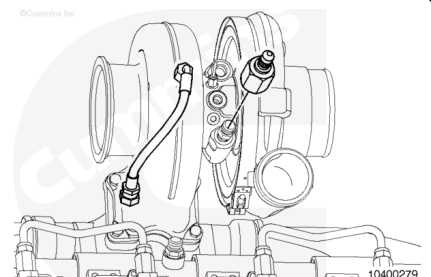
If the pressure does decrease and the hose connections are **not** leaking, replace the turbocharger.



Remove the pressure regulator.

Remove the coolant return reducing adapter.

Install the coolant return line. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html>)

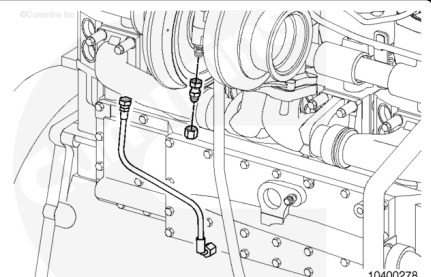


Remove the cap from the coolant supply adapter.

Remove the adapter from the coolant supply fitting.

Install the coolant supply hose or line. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html>)

Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)

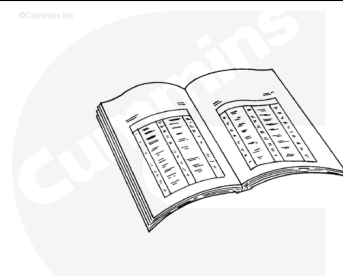


Preparatory Steps

All Applications Except Marine and Rear Mount Rail

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not rescued, dispose of in accordance with local environmental regulations.

Note : For marine applications use the following procedure. Refer to Procedure 010-037 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-037-tr.html>)

Mechanically actuated injectors

- Remove the intake and exhaust pipes from the turbocharger.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the turbocharger heat shield, if installed. Refer to Procedure 010-076 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html>)
- Remove the air crossover pipe. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>)
- Remove the oil supply line. Refer to Procedure 010-046 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-046.html>)
- Remove the oil drain line. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

- Remove the coolant supply and return hoses. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html)
- Inspect the charge air cooler for debris. Refer to Procedure 010-027 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-027.html)

Electronically actuated injectors

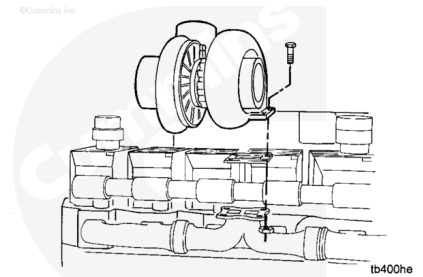
- Disconnect the turbocharger speed sensor from the wiring harness and remove the clamp securing the speed sensor wire. Use the following procedure in Section 19 of the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System Series Engine, Bulletin 4021493. Refer to Procedure 019-390 in Section 19.
(/qs3/pubsys2/xml/en/procedures/13/13-019-390.html)
- Remove the intake and exhaust pipes from the turbocharger.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Remove the turbocharger heat shield, if installed. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html)
- Remove the air crossover pipe. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)
- Remove the oil supply line. Refer to Procedure 010-046 in Section 10 (/qs3/pubsys2/xml/en/procedures/20/20-010-046.html)
- Remove the oil drain line. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Remove the coolant supply and return hoses. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html)
- Inspect the charge air cooler for debris. Refer to Procedure 010-027 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-027.html)

Remove

All Applications Except Marine and Rear Mount Rail

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



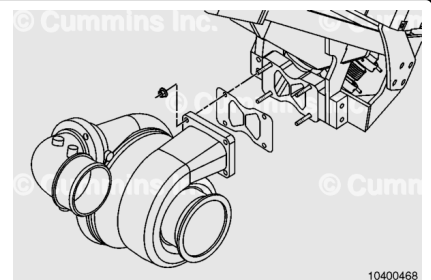
Remove the mounting capscrews, nuts, and turbocharger.

Remove and discard the gasket.

Rear Mount Rail Applications

Remove the mounting nuts and the turbocharger.

Remove the turbocharger and discard the gasket.

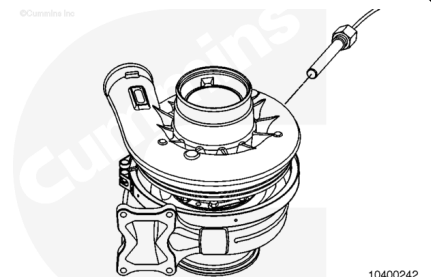


Disassemble

Note : Electronically actuated injector engines are equipped with a turbocharger speed sensor.

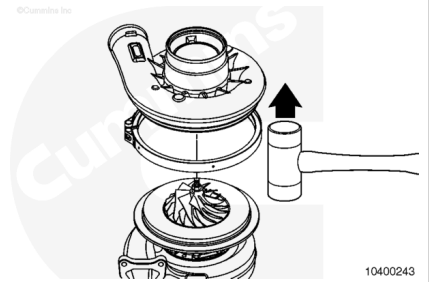
Remove the speed sensor from the turbocharger, if equipped. Use the following procedure in Section 19 of the Troubleshooting and Repair Manual Electronic Control System QSK19 CM850 Modular Common Rail System Series Engine, Bulletin 4021493. Refer to Procedure 019-390 in Section 19. (</qs3/pubsys2/xml/en/procedures/13/13-019-390.html>)

Loosen and remove the v-band clamp securing the compressor housing.



⚠ CAUTION ⚠

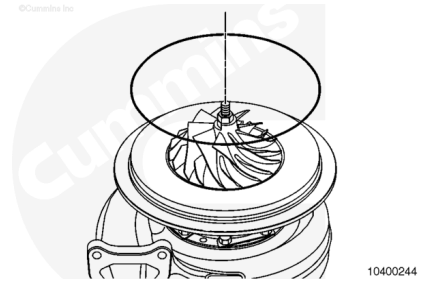
The compressor blades can be easily damaged with the compressor cover removed.



Note : Turbocharger HX83 uses a titanium compressor wheel. The edge of the wheel is very sharp. Handle the compressor wheel with care.

Use a mallet to gently tap the compressor cover to remove the cover from the turbocharger.

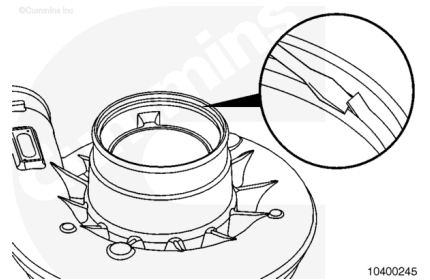
Remove the o-ring and discard.



⚠ CAUTION ⚠

The impeller blades can be easily damaged when the compressor cover is removed.

If the compressor cover has an inlet baffle, it can be removed with a flat tip screwdriver. Pry the retaining ring up and remove the retaining ring.

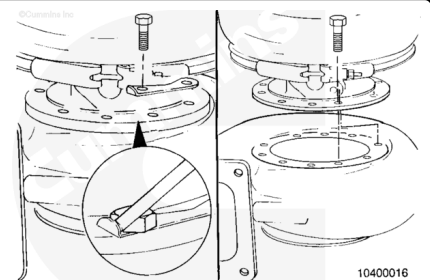


⚠ CAUTION ⚠

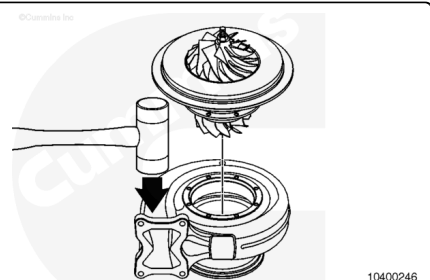
The turbine blades can be easily damaged when the turbine housing is removed.

Remove the turbine housing by bending the lock plate tabs away from the capscrew heads.

Remove the capscrews.

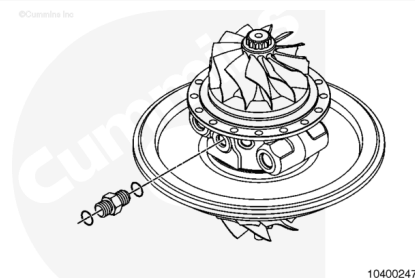


Make sure the turbine wheel rotates freely inside the turbine housing. Tap on the turbine housing with a mallet to separate the turbine housing from the bearing housing.



Remove the oil supply fittings and the coolant lines from the center housing assembly.

Discard the o-rings.



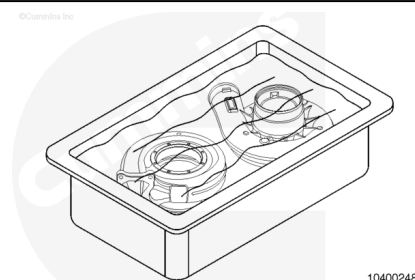
Clean and Inspect for Reuse

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

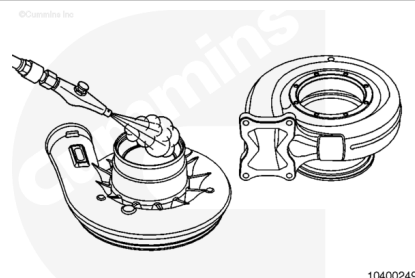
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the compressor cover and turbine housing in parts cleaner to loosen deposits.

⚠ WARNING ⚠

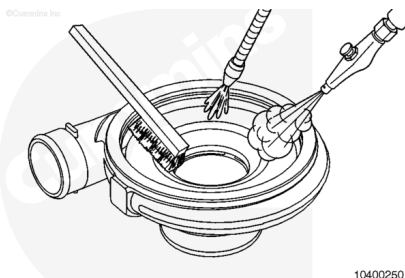
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Dry the compressor housing with compressed air.

Heavy scale deposits can be removed using a soft bristle brush.

Wash and dry the parts after brushing.



⚠ CAUTION ⚠

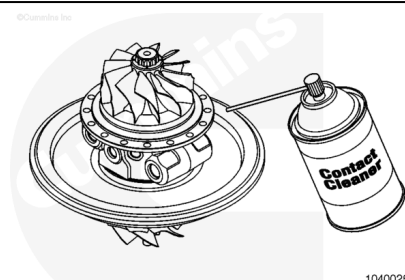
Do not use compressed air to dry or spin the turbine wheel as this can cause damage to the turbocharger bearing system.

If chemical and brush cleaning is **not** effective, bead blast with plastic abrasive media, Part Number 3822735, or equivalent.

Wash and dry parts after bead blasting.

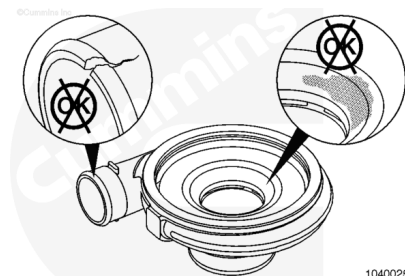
The center housing rotating assembly can be cleaned with electrical contact cleaner, Part Number 3824510, or equivalent.

Do **not** spray contact cleaner into the oil and coolant ports.



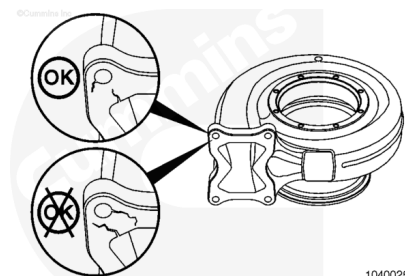
Inspect the compressor cover for cracks, dents, and other damage.

Inspect the inside of the compressor cover for signs of scoring or damage from the compressor wheel. Replace the compressor cover, if damaged.

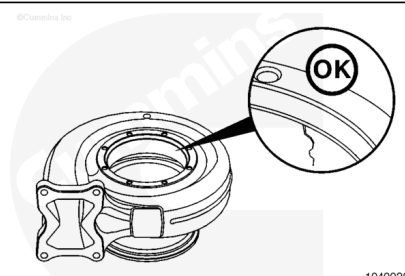


Inspect the turbine housing for cracks and other damage. Cracks larger than 10 mm [0.39 in] can cause an exhaust leak. Replace the turbine housing if damage is found.

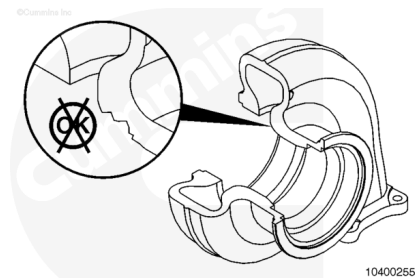
Follow the criteria shown in the illustration.



Minor cracks in the turbine housing internal wall, where exhaust gas flow enters the turbine wheel, is an acceptable condition. The turbine housing can be reused.



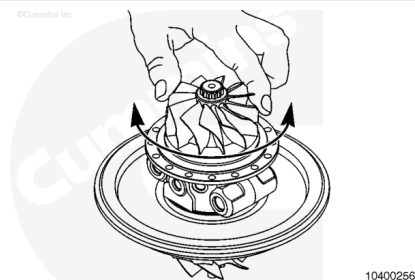
Inspect the turbine housing for signs of contact with the turbine wheel. Damaged turbine housings **must** be replaced.



Inspect the center housing rotating assembly for any signs of cracks, damaged threads, or other damage.

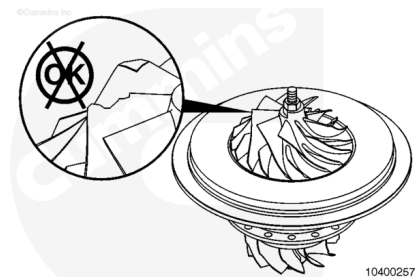
Rotate the turbine wheel by hand to make sure it spins freely.

Replace the center housing rotating assembly if it is damaged or does **not** spin freely.



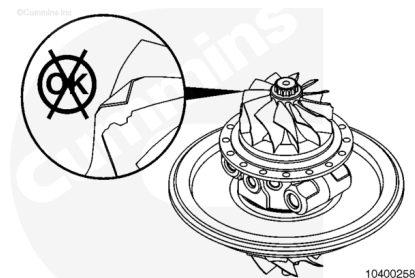
Inspect the compressor wheel for damaged, broken, or bent blades.

Do **not** attempt to repair or straighten the blades; instead replace the center housing rotating assembly.



Inspect the turbine wheel for damaged, broken, or bent blades.

Do **not** attempt to repair or straighten the blades; instead replace the center housing rotating assembly.



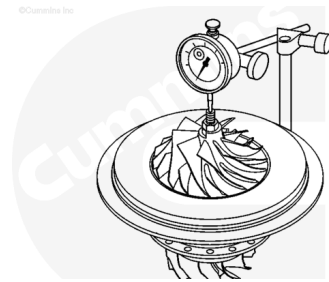
Measure

Measure the end play with a dial indicator.

Replace the center housing rotating assembly if the end play is **not** within specification:

Measurements

	mm	in
End Play	0.025 to 0.152	0.001 to 0.006



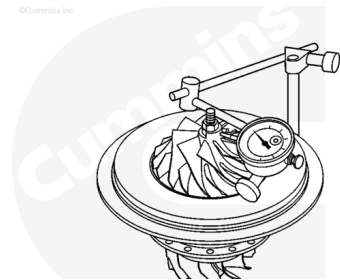
10400259

Measure the movement with a dial indicator.

Replace the center housing rotating assembly if the radial movement is **not** within specification:

Measurements

	mm	in
Radial Movement	0.254 to 0.787	0.010 to 0.031



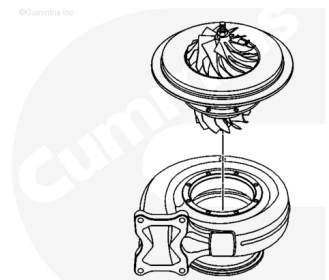
10400260

Assemble

⚠ CAUTION ⚠

The turbine blades can be easily damaged when installing the assembly in the turbine housing.

Assemble the center housing rotating assembly on to the turbine housing.

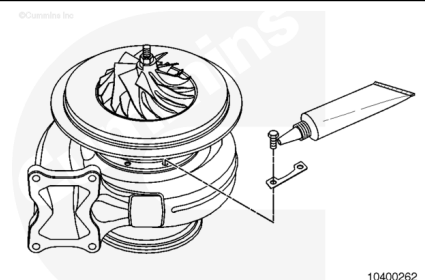


10400261

Lubricate the threads of the new capscrews with anti-seize compound, Part Number 3824732, or equivalent.

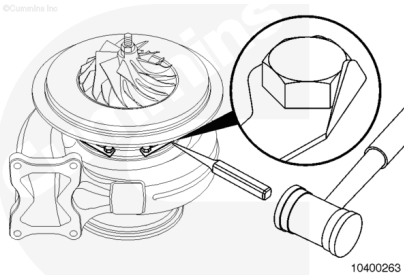
Install the capscrews, with new lock plates, to the turbocharger turbine housing.

Torque Value: 20.3 n·m [180 in-lb]



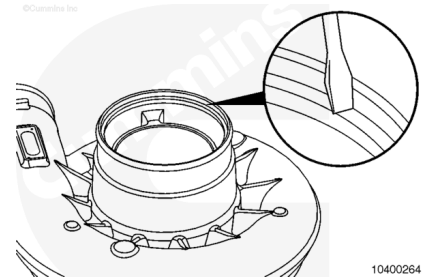
10400262

Use a hammer and punch to fold the locking plate onto the flat of the capscrew to prevent loosening.



10400263

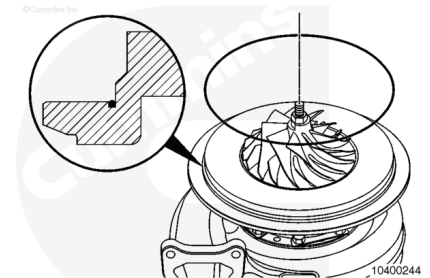
Assemble the inlet baffle into the compressor cover and secure with a retaining ring. Hold one end of the retaining ring in the groove. Press the remainder of the retaining ring into position. Use a flat tip screwdriver to make sure the retaining ring is correctly seated in the compressor cover groove.



10400264

Note : Turbocharger HX83 uses a titanium compressor wheel. The edge of the wheel is very sharp. Handle the compressor wheel with care.

Assemble the compressor housing new o-ring on the turbocharger.

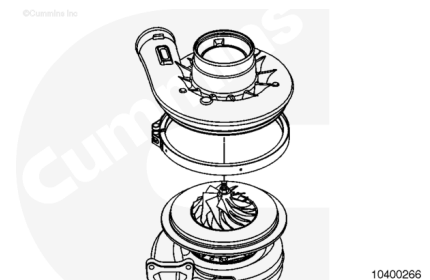


10400244

⚠ CAUTION ⚠

Do not damage the compressor wheel blades during assembly.

Assemble the v-band clamp and compressor cover onto the turbocharger.

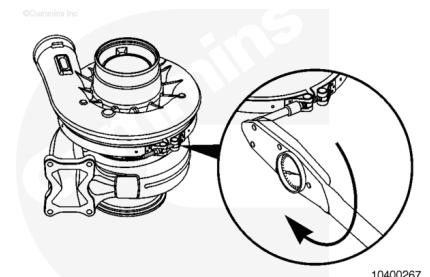


10400266

Place the v-band clamp in position.

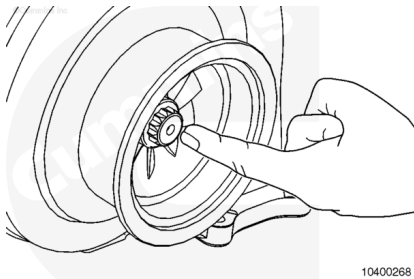
Tighten the v-band clamp.

Torque Value: 12 n•m [106 in-lb]



10400267

Make sure the compressor wheel and turbine wheel rotate freely.

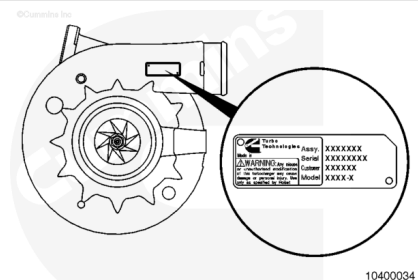


Install

All Applications Except Marine and Rear Mount Rail

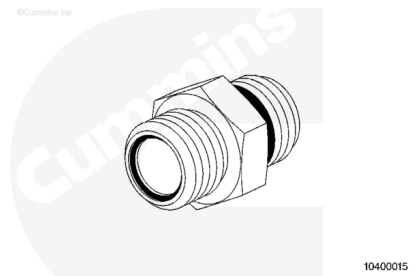
Cummins Turbo Technologies™ turbocharger(s) are used on the QSK19 engine. See the data tag on the turbocharger to determine the model.

The name is also cast on the housing of a turbocharger.



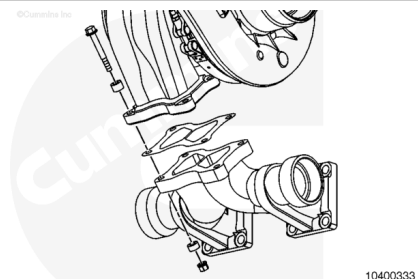
Note : All turbocharger fittings are the flat face o-ring type fittings. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

The Cummins Turbo Technologies™ turbocharger uses a 9/16-18 UNF, straight thread, o-ring type oil supply fitting.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply an anti-seize compound to the mounting capscrews.

Install the gasket.

Note : The word OUT or raised gasket bead must be positioned toward the turbocharger.

For engines with mechanically and electronically actuated injectors, install the turbocharger, turbocharger capscrews, nuts, and spacers. The 12.7 mm [1/2 inch] spacer **must** be on the turbocharger side of the joint, while the 9.525 mm [3/8 inch] spacer **must** be on the manifold side or the bolted joint.

Tighten the turbocharger capscrews using sequence and torque steps below.

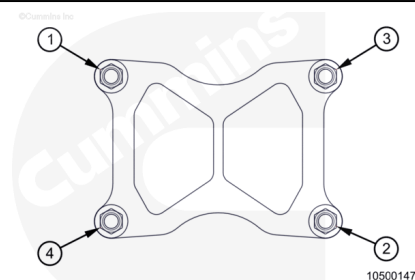
Torque Value:

Step 1: 20 n•m [177 in-lb]

Torque Value:

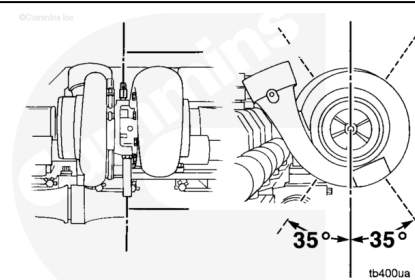
Step 2: 47 n•m [35 ft-lb]

Perform the final torque value in Step 2 using the sequence shown a second time.



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

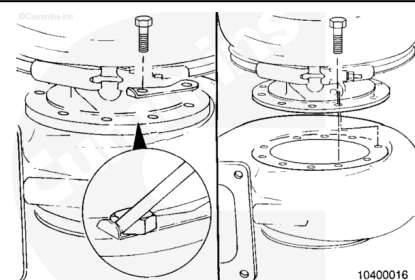
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



For bearing housing adjustment, bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

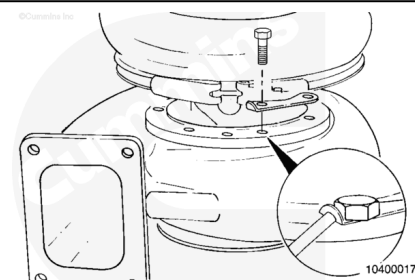


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]

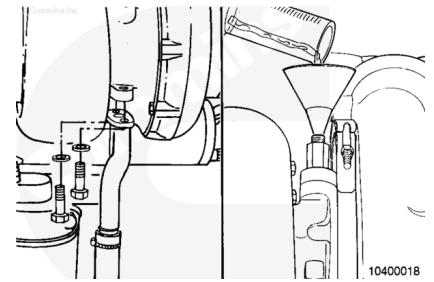
Bend the lockplate tabs over the capscrews.



Install the oil drain line. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)

Pour clean engine oil into the turbocharger oil supply fitting.

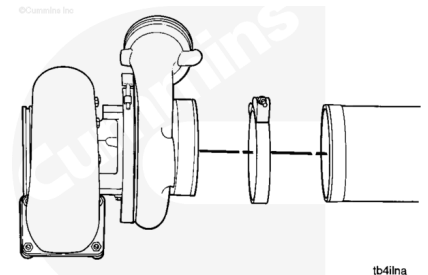
Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

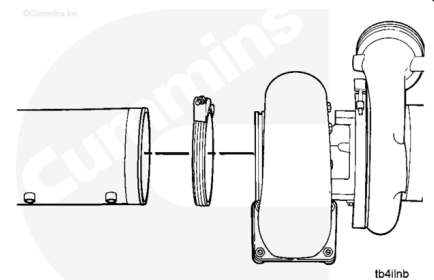
Torque Value: 8 n•m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

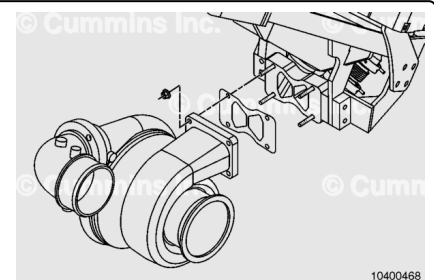
Torque Value: 8 n•m [71 in-lb]



Rear Mount Rail Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply an anti-seize compound to the mounting studs.

Install the gasket on the studs. The word **OUT** **must** be positioned toward the turbocharger.

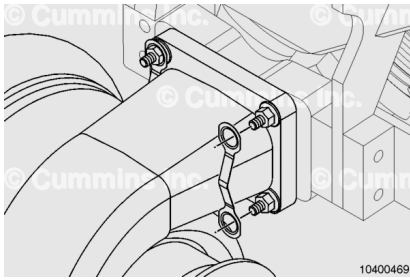
Install the turbocharger.

Install and finger tighten the nuts on the studs.

Tighten the mounting nuts in a criss-cross pattern.

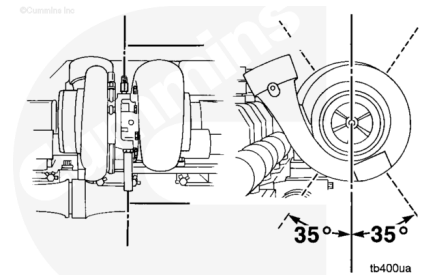
Torque Value: 57 n•m [42 ft-lb]

Install the two lockplates over the turbocharger mounting nuts.



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

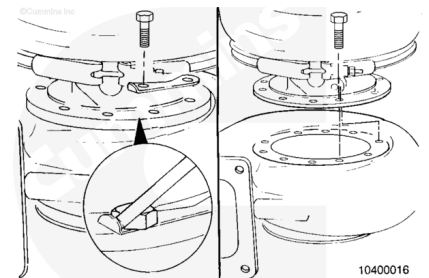
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



For bearing housing adjustment, bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

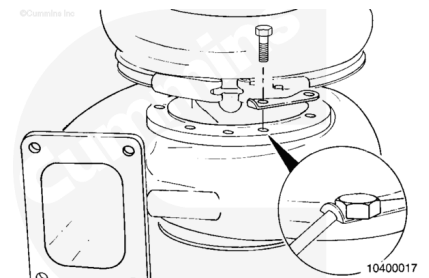


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n·m [177 in-lb]

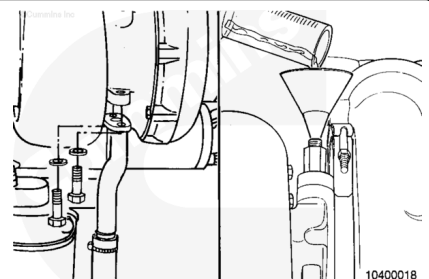
Bend the lockplate tabs over the capscrews.



Install the oil drain line. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

Pour clean engine oil into the turbocharger oil supply fitting.

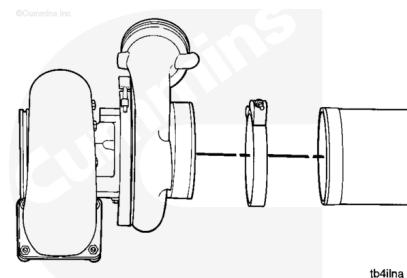
Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

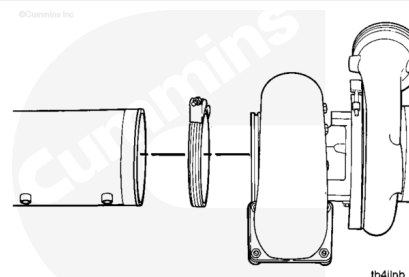
Torque Value: 8 n•m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

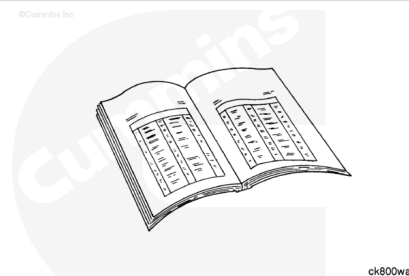
Torque Value: 8 n•m [71 in-lb]



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



- For engines with electronically actuated injectors, connect the turbocharger speed sensor to the wiring harness and install the clamp securing the speed sensor wire. Use the following procedure in Section 19 of the Troubleshooting and Repair Manual Electronic Control System QSK19 CM850 Modular Common Rail System Series Engine, Bulletin 4021493. Refer to Procedure 019-390 in Section 19. (</qs3/pubsys2/xml/en/procedures/13/13-019-390.html>)
- Clean debris from the charge air cooler if debris is present. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-027.html>)
- Install the oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-046.html>)
- Install the coolant supply and return hoses. Refer to Procedure 010-041 in Section 10.

(/qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html)

- Install the air crossover pipe. Refer to Procedure 010-019 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)
- Install the turbocharger heat shield, if installed. Refer to Procedure 010-076 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Operate the engine at 71°C [160°F] coolant temperature and check for leaks.